

SUNNI KUMAR

+91 6394540805 • contact.sunnyarya@gmail.com • linkedin.com/in/sunni-kumar • github.com/sunny-arya-codes

SUMMARY

Backend / GenAI Engineer with production experience at FreshMenu, pursuing B.S. in Data Science at IIT Madras. Building scalable backend systems (FastAPI, Spring Boot) and production-grade AI solutions (RAG pipelines, LLM integrations). Focused on high-performance APIs, event-driven architectures, and intelligent automation systems. Strong foundation in data structures, algorithms, and machine learning with hands-on experience in full-stack development.

EXPERIENCE

FRESHMENU

Software Developer Engineer

Bengaluru, Karnataka

(Mar 2024 – Present)

Automation & Financial Reconciliation Platform (AI-Enabled)

- Built **automated financial reconciliation workflows** to detect order and settlement mismatches across multiple third-party platforms (**Zomato, Swiggy, Magicpin, Bitsila**) by comparing partner-reported data with internal records.
- Developed a finance-facing **reconciliation dashboard** using **FastAPI, Next.js, and SQL**, providing clear visibility into mismatches through **scheduled daily reconciliation jobs** (cron-based).
- Reduced significant manual reconciliation effort by automating mismatch detection and reporting, **saving approximately 15+ hours per week** for the finance team.
- Designed a **configurable reconciliation rules engine** by vectorizing partner-specific agreement PDFs using **FAISS** and integrating an **OSS GPT-based LLM**, enabling finance teams to manage rules dynamically without code changes.

Procurement Automation System (AI-Driven)

- Built an **end-to-end procurement automation workflow** connecting kitchen staff, procurement teams, and vendors to streamline **PO creation, approval, and invoice processing**.
- Automated **vendor notifications via email** after PO approvals and handled invoice intake through **email-based workflows**.
- Implemented **invoice data extraction** using **OpenCV**, structuring extracted information to match existing database schemas.
- Enabled **automatic PO-invoice matching** using **FastAPI** and an **NVIDIA-hosted OSS GPT-based LLM**, eliminating manual reconciliation and reducing operational delays.

Backend Engineering & ONDC Integrations

- Worked on an existing **Java & Spring Boot-based core ordering system**, fixing multiple API bugs, improving performance, and **optimizing response times** for production workloads.
- Improved **API reliability under high traffic** by resolving timeout issues, null-pointer errors, and strengthening **application-level error handling** across local and production environments.
- Designed and developed **new REST APIs** to enable addon and variant features, owning complete business logic and **seamless integration** within the existing ordering flow.
- Led **end-to-end ONDC third-party integrations** with **Magicpin and Bitsila**, independently coordinating with partner teams and implementing **secure JWT-based authentication**.
- Built **asynchronous order processing** using **ActiveMQ**, including webhooks and background jobs, with **custom retry logic** to prevent order loss during transient failures.
- Actively handled **production issues**, monitored logs using **Kibana**, and configured **email alerts** to ensure system stability under high load.

MenuHR (HRMS Platform)

- Developed a **REST API-based HRMS application** for **employee management, roster management, and attendance tracking** (clock-in/clock-out).
- Built backend services using **Java, Spring Boot, and MySQL** to manage employee records, shifts, and attendance logs.
- Implemented a **web interface using Next.js** to support daily HR and workforce operations.

Technologies: Python, Java, Spring Boot, FastAPI, Next.js, MySQL, PostgreSQL, SQL, REST APIs, JWT Authentication, ActiveMQ, Webhooks, OpenCV, FAISS (Vector DB), OSS GPT-based LLM, Docker, Git, Kibana

FRESHMENU

Software Developer Intern

Bengaluru, Karnataka

(Sep 2023 – Mar 2024)

- Spent initial phase understanding the existing **Java & Spring Boot-based ordering platform**, including system architecture, service interactions, and data flow across core modules.

- Worked on **bug fixes and minor enhancements** in existing **REST APIs**, gaining hands-on experience with production-grade backend codebases.
- Implemented a **third-party channel integration (Thrive)**, similar to Magicpin, contributing to order flow integration and **API-level communication** under guidance.
- Assisted in implementing **request validation, error handling, and basic API documentation** for internal and external consumers.
- Collaborated with senior engineers during **code reviews, sprint planning, and deployments**, learning development best practices in a startup environment.
- Gained exposure to **production debugging, log analysis, and incident resolution** workflows while supporting live systems.

Technologies: Java, Spring Boot, PostgreSQL, SQL, REST APIs, Git, Linux, Docker

KEY PROJECTS

AI AGENT FOR ACADEMIC GUIDANCE

Jan 2025 – Mar 2025

- Engineered full-stack **RAG (Retrieval-Augmented Generation) pipeline** for intelligent document search and context-aware question answering.
- Built FastAPI backend with **JWT authentication**, role-based access control, and rate limiting for secure API access.
- Implemented vector similarity search using pgvector extension with configurable retrieval thresholds and reranking strategies.
- Developed document chunking strategies, embedding generation with OpenAI APIs, and optimized semantic retrieval for accuracy.
- Created Vue 3 frontend with responsive design, real-time chat interface, and document upload functionality.
- Deployed production infrastructure on **AWS EC2** with Nginx reverse proxy, SSL certificates, and automated CI/CD pipelines via GitHub Actions.

Technologies: Python, FastAPI, LangChain, pgvector, PostgreSQL, OpenAI API, Vue.js 3, AWS EC2, Vercel, Docker, Nginx

SPATIO-TEMPORAL CRIME PREDICTION

June 2024 – Aug 2024

- Developed **multi-class classification model** using XGBoost to predict crime categories by analyzing complex spatio-temporal patterns.
- Performed comprehensive **exploratory data analysis (EDA)** on high-dimensional datasets with 800K+ crime records across multiple years.
- Engineered advanced features including time-based (hour, day, month, seasonality), location-based (district, beat, ward), and cyclical encodings.
- Implemented k-fold cross-validation, hyperparameter tuning with GridSearchCV, and comprehensive model evaluation with precision, recall, and F1.
- Created interactive visualizations using Matplotlib and Seaborn to communicate insights on crime hotspots and temporal patterns.

Technologies: Python, XGBoost, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebooks

EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Chennai

B.S. in Data Science and Applications

(May 2022 – Apr 2026)

Coursework: Machine Learning, Deep Learning, Database Systems, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Python Programming, Business Data Management

MASAI SCHOOL

Full Stack Development

Intensive Backend-Focused Program (9-9-6 Curriculum)

(Apr 2022 – Dec 2022)

Rigorous practical full-stack training with focus on **Java, Spring Boot, and MySQL**. Solved **1200+ DSA problems** across platforms. 12 hrs/day, 6 days/week immersive learning with real-world project experience, code reviews, and industry-standard development practices.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML, CSS
GenAI & ML: LangChain, RAG Pipelines, pgvector, OpenAI API, Hugging Face, PyTorch, XGBoost, Scikit-learn, Pandas, NumPy

Backend & Frameworks: Spring Boot, FastAPI, Node.js, Express.js, REST APIs, GraphQL, OAuth 2.0, JWT

Databases: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases (pgvector)

DevOps & Cloud: Docker, Git, GitHub Actions, AWS (EC2, S3, RDS), Linux, Nginx, CI/CD Pipelines

Tools & Practices: Swagger, Postman, JUnit, Mockito, Agile/Scrum, Code Reviews, Technical Documentation